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NeuroSEED: A Neuromorphic Scalable Event-Driven SoC with Direct Multiplexing for Spike Detection and Wireless Telemetry

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***Abstract*—Implantable multi-channel neural interfaces are essential for high-resolution, long-term brain-machine interface applications, yet conventional designs are constrained by power consumption, data bandwidth, and silicon area. We present NeuroSEED, a 32-channel neuromorphic scalable neural interface inspired by biological neuron signaling. This system introduces four key innovations to address the bottlenecks of scalability and efficiency: (1) an area-efficient time-division multiplexed analog front-end ($0.0032\ mm^2/channel$) with a dedicated digital-servo electrode DC offset cancellation loop; (2) an analog-to-spike converter (ASC) that employs adaptive event-driven sampling; (3) a clockless spike detector utilizing valley-to-peak detection for robust spike identification; and (4) an event-triggered impulse-radio ultra-wideband (IR-UWB) transmitter for energy-efficient wireless spike telemetry. Fabricated in 40-nm CMOS, NeuroSEED achieves 1.38 μW per recording channel and 4.6 μW transmitter power, with more than 500$\times$ data reduction compared to Nyquist-rate sampling. Experimental results, validated with in-vivo neural datasets, confirm low noise, compact area, and robust system-level operation, positioning NeuroSEED as a highly efficient SoC for high-density neural recording.**

***Index Terms*—Neuromorphic systems, neural interface, compressive sampling, analog-to-spike converter.**

I. Introduction

THE channel count of implantable brain-machine interfaces (iBMIs) has grown rapidly due to the increasing complexity of brain-related applications, which now range from basic brain disorder monitoring and motor control to high-fidelity speech and sensory decoding [1]–[4]. While Moore's Law has significantly enhanced digital circuit performance, the benefits of CMOS process scaling for analog circuits remain limited due to fundamental constraints such as thermal noise, mismatch, and parasitic effects. Consequently, further reductions in power and area for neural signal acquisition front ends necessitate innovative circuit architectures that enable fine-grained neural activity recording with scalability while meeting stringent power and area constraints. Additionally, reducing the volume of wirelessly transmitted data without compromising task performance is essential, as the power consumption associated with data transmission increases significantly with the number of channels. This rise in energy demand directly challenges the thermal safety of the implantable chip.

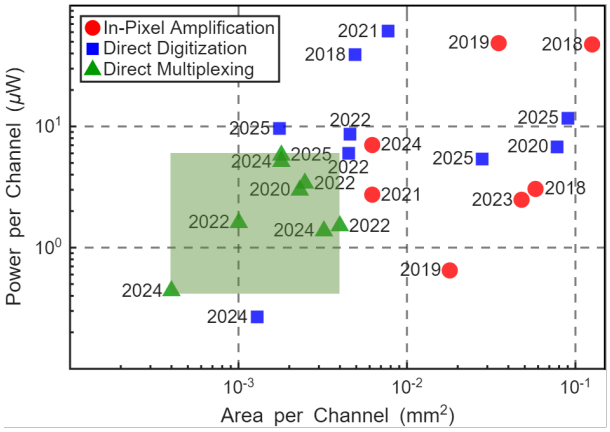


Fig. 1: Power and area trends of neural signal acquisition systems from 2018 to 2024 [5]–[29]. Shaded regions highlight the performance ranges of the Direct Multiplexing architecture.

In recent years, significant research efforts have been directed toward minimizing the area and power consumption of neural signal acquisition front-ends. These efforts have primarily focused on three key architectural approaches:

- **In-Pixel Amplification [5]–[11]**: Each channel is equipped with dedicated acquisition and amplification circuitry, enabling continuous optimization of individual channel performance.
- **Direct Digitization [12]–[17], [26], [28], [29]**: The integration of acquisition and amplification modules with quantization circuits allows for direct digital signal output, eliminating the need for intermediate analog processing stages.
- **Direct Multiplexing [18]–[25], [27]**: Signals from multiple electrodes are sequentially routed to a shared set of high-bandwidth acquisition circuits, enabling efficient utilization of a single acquisition module across multiple electrodes.

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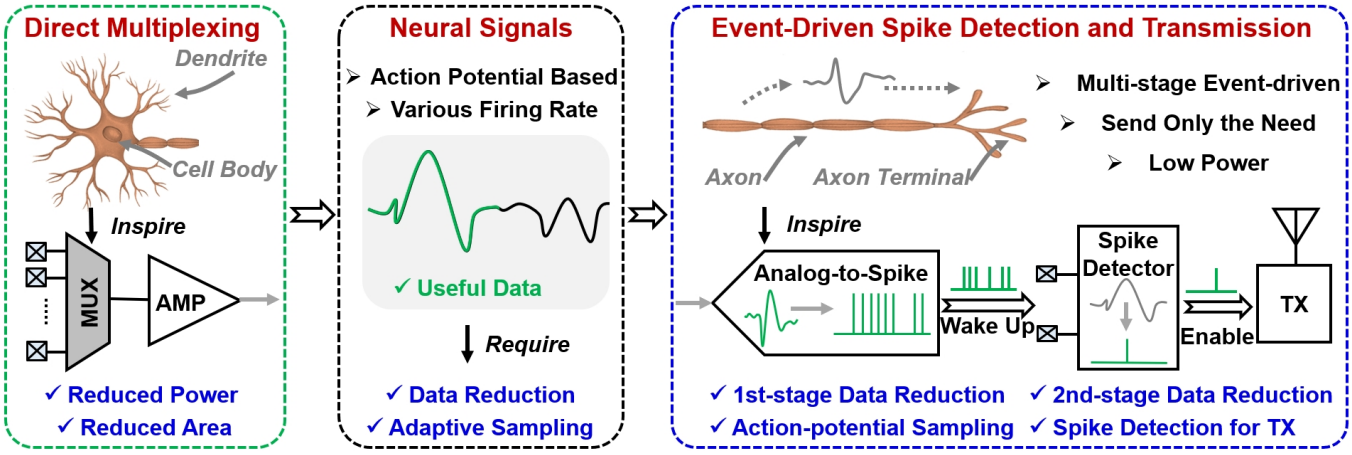


Fig. 2: System concept of the proposed neuromorphic neural interface.

Furthermore, the integration of application-specific functionalities into these architectures enhances system compactness by ensuring that only relevant signals are processed from the outset, thereby reducing redundancy and improving overall efficiency. Fig. 1 summarizes the power and area trends of representative works from 2018 to 2025 [5]–[29], with shaded regions highlighting the significant potential and ongoing advancements of the direct multiplexing architecture. While this approach significantly reduces the silicon footprint, it introduces additional design challenges, particularly in mitigating inter-channel DC offsets and in-band integrated noise. Moreover, the increased switching activity and clocking overhead pose challenges in noise management, signal integrity, and power efficiency.

Despite the substantial reduction in average area per channel, power consumption remains a critical concern, especially for large-scale neural interfaces. As these systems scale to thousands of channels, the need for continuous neural signal acquisition and wireless data transmission leads to significant power dissipation. Conventional architectures face fundamental trade-offs among resolution, bandwidth, and energy efficiency, making it challenging to meet the stringent power constraints required for chronic implantation.

Inspired by the efficiency of biological neurons, this paper presents NeuroSEED, a neuromorphic neural interface. Expanding upon the framework in [23], this work realizes the complete NeuroSEED system, adding a rigorous theoretical analysis of the direct-multiplexing architecture with offset cancellation, a co-integrated low-power IR-UWB transmitter, and comprehensive system-level validation. By emulating the high fan-in of biological neurons, the design multiplexes the entire readout chain, achieving a compact AFE footprint of 0.0032 mm^2 per channel even with integrated electrode DC offset (EDO) cancellation. NeuroSEED employs neuromorphic principles across sampling, processing, and transmission, encoding information as event-driven spikes [30], [31]. An analog-to-spike converter (ASC) generates digital spike trains via adaptive sampling, significantly reducing power. A subsequent asynchronous spike detector, triggered by the ASC, eliminates clocking power and compresses data by over 500×. Finally, the compressed spikes selectively activate the transmitter. This architecture processes high-volume neural data with a power consumption of only 1.38 μW per channel, offering a highly efficient solution for neural interfaces.

Recent 2025 state-of-the-art works converge on two complementary directions: system-level closed-loop SoCs with on-chip spike sorting or edge-AI inference [32], [33], and circuit-level ultra-low-NEF front-ends based on continuous-time $\Delta\Sigma$ [28], [34], TDM+SAR [27], and OTA-C/boxcar architectures [17], [26]. NeuroSEED is, to our knowledge, the first TDM neural SoC that co-integrates a chopper AFE with a *digital-domain* offset servo, an event-driven ASC with clockless valley-to-peak detection, and a spike-gated IR-UWB telemetry path, yielding >500× on-chip data reduction and transmitter-side power that scales linearly with the firing rate.

The remaining sections of the paper are arranged as follows. Section II presents the overall architecture of the proposed NeuroSEED system, detailing its 32-channel direct-multiplexing front-end and event-driven processing pipeline. Section III-A describes the design considerations and circuit implementation of the analog front-end. Section III-B explains the analog-to-spike converter and the clockless spike detection mechanism, which perform neuromorphic data compression. The digital feedback loop for mitigating electrode DC offset is detailed in Section III-C. Section III-D presents the event-triggered impulse-radio ultra-wideband (IR-UWB) transmitter. Section IV reports comprehensive measurement results from the fabricated chip, including validation with an in-vivo dataset and a comparison with state-of-the-art works. Finally, Section V concludes the paper.

II. NeuroSEED Architecture

Figure 3 illustrates the architecture of the proposed 32-channel neuromorphic neural interface. The 32 channels are partitioned into eight groups, each sequentially connected to a 4-channel multiplexing chopper to minimize the number of required chopper switches [20]. The multiplexed neural signals, modulated by chopping, are amplified by a capacitively coupled AFE, and digitized by an analog-to-spike conversion module with cooperative offset-cancellation feedback, which dynamically reduces the electrode DC offset.

Following multichannel neural recording, the proposed NeuroSEED system performs signal sampling, processing, and

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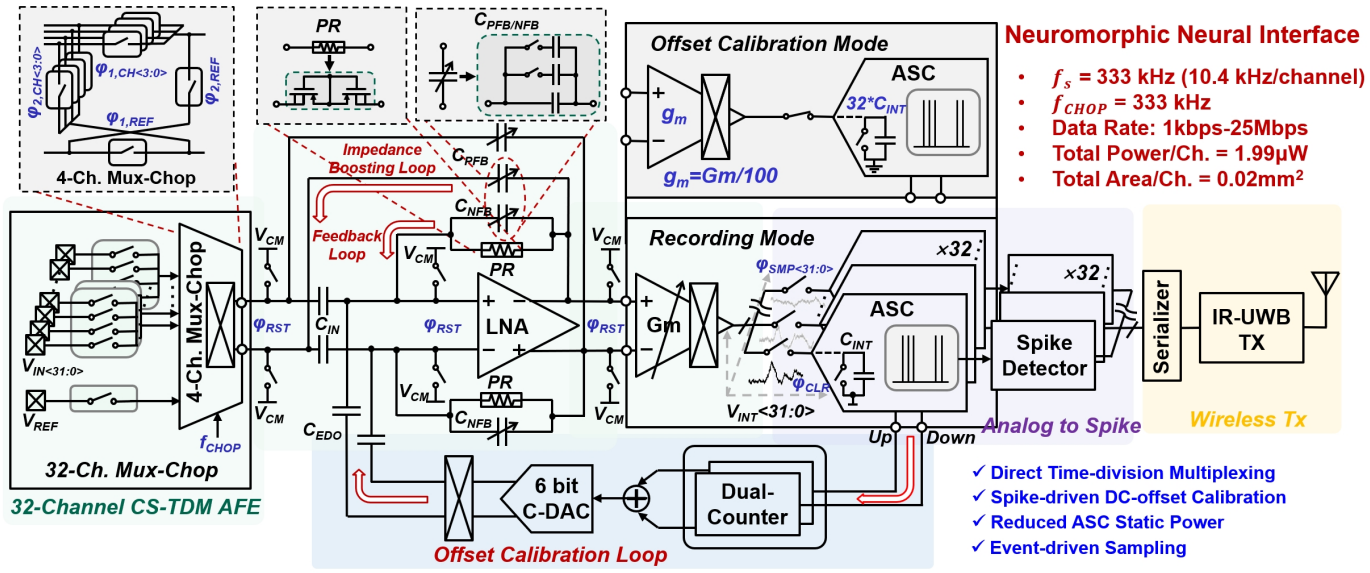


Fig. 3: Overall block diagram of the proposed system, implementing a 32-channel direct-multiplexing front-end combined with event-driven processing.

transmission based on neuromorphic principles, where information is encoded and processed as event-driven digital spikes. The ASC generates a spike train reflecting neural activity, employing adaptive sampling aligned with neuron firing rates to minimize redundant acquisition and reduce power consumption. This spike output triggers a clockless spike detector, which further compresses the signal through asynchronous, event-based detection, achieving over 500× data reduction compared to conventional continuous sampling. The compressed spike data selectively activates the wireless transmitter, enabling efficient transmission only when relevant neural events occur. This architecture allows the NeuroSEED to process large-scale neural data with both high energy efficiency and compact area utilization.

Neural signals are sampled at 10.4 kHz per channel through the control of input switches, with a chopping frequency of 333 kHz to suppress electronic offset and 1/f noise. While high-fidelity spike sorting often utilizes sampling rates between 20–30 kHz to capture fine waveform features, a rate of 10.4 kHz exceeds the Nyquist rate for the relevant neural action potential bandwidth (typically < 5 kHz). This rate is sufficient for reliable spike detection and coarse classification, which is the primary objective of this ultra-low-power interface. The selection of 10.4 kHz represents a deliberate trade-off to minimize dynamic power consumption in the multiplexing and serialization stages.

A quantitative comparison of this sampling choice against a 25 kHz baseline and an analysis of scaling behaviour toward higher channel counts are deferred to Section III-A, where the required front-end bandwidth model, the in-silico matched-spike study, and the supporting figures are introduced.

III. NeuroSEED Circuit Implementation

This section details the circuit-level implementation of the NeuroSEED interface, following the signal path from the area-efficient, 32-channel direct multiplexing AFE to the event-triggered IR-UWB transmitter. The discussion focuses on the key design aspects of each stage: first, the AFE techniques for mitigating the noise and large DC offsets inherent in time-division systems, the latter addressed by a dedicated feedback loop. Subsequently, we detail the neuromorphic core—comprising an ASC and spike detector—which achieves significant data compression prior to the final, power-efficient wireless transmission stage.

A. 32-Channel Direct Multiplexing Front-End

Direct multiplexing is an effective technique for reducing the area and power consumption of acquisition front-ends, as it enables rapid sampling across multiple electrode sites using a single front-end circuit. The circuit area scales inversely with the multiplexing ratio N, while the power consumption scales with the bandwidth, which increases by at least N-fold. Despite the higher bandwidth requirement, sharing auxiliary modules across channels helps maintain relatively low average power consumption, particularly when scaling to a large number of channels.

However, several challenges remain in direct multiplexing schemes. First, electrode offset voltages (EOVs) up to $\pm 50\,\mathrm{mV}$ can saturate the AFE when coupled through the multiplexer [21], [35], [36], necessitating area-intensive compensation circuits such as low-pass filers (LPFs), MUX/DEMUX structures, and LMS adaptive filters to extract, process, and cancel EOVs. Conventional AC coupling and DC-servo loops, effective in per-channel AFEs, are unsuitable for multiplexed designs. Second, multiplexing N channels requires an amplifier bandwidth that is N times larger, which forces the use of smaller input differential pairs and increases the $1/f$ noise corner, while also elevating integrated thermal noise due to the large bandwidth. Third, post-MUX chopping further degrades the input impedance compared to multiplexing alone. Consequently, many direct multiplexing neural recording systems adopt compromises, such as limiting the recording bandwidth (e.g., 500 Hz), increasing circuit area, or tolerating a higher

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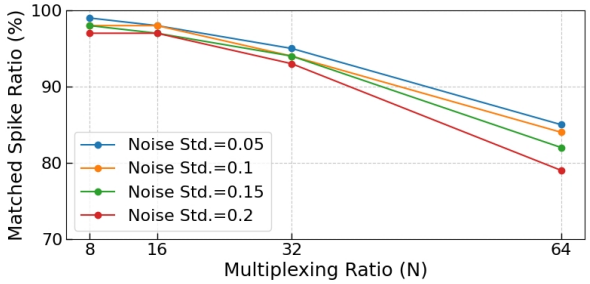


Fig. 4: Matched spike ratio vs. multiplexing ratio (N) under different noise distributions (Std.=0.05,0.1,0.15,0.2).

noise efficiency factor (NEF), thereby underscoring the need for advanced circuit solutions.

1) Design Consideration: Due to rapid electrode switching and variations in tissue equilibrium potential, the multiplexed voltage appears as a step input. For a single-pole low-pass filter response, the required bandwidth to achieve a given transient settling accuracy is expressed as

$$f_{BW} \geq \frac{-\ln \epsilon}{2\pi T_s}, \tag{1}$$

where ϵ is the fractional settling error (0.1% for 10-bit accuracy), and T_s is the available settling time, determined by the per-channel sampling rate f_{ch} and the multiplexing ratio N as $T_s = 1/(f_{ch} \cdot N)$. Increasing N reduces T_s, thereby requiring a larger bandwidth to achieve the same settling error. While a larger bandwidth reduces ϵ, it also increases both power consumption and in-band noise. Hence, selecting an appropriate bandwidth and optimal N is critical to balancing performance and efficiency.

To guide the front-end design, in-silico modeling was performed using a simulated dataset [37] sampled at 24 kHz with annotated neural spike locations. To minimize power consumption, the AFE bandwidth was fixed at 700 kHz with a closed-loop gain of 34 dB. A smaller bandwidth permits larger input differential pair devices, thereby reducing flicker noise. Four 10-second data segments with noise standard deviations of 0.05, 0.1, 0.15, and 0.2 were simulated for multiplexing ratios N = 8, 16, 32, and 64, with the per-channel sampling rate fixed at 10.4 kHz. Fig. 4 shows the matched spike ratio for different N and noise levels, indicating that $N = 64$ significantly degrades spike detectability. Based on these results, N = 32 was selected for the front-end design, as it maintains a matched spike ratio above 93% across all noise levels while achieving an optimal balance between spike fidelity and power efficiency.

Compared to 20–30 kHz systems, the 10.4 kHz per-channel rate introduces only modest peak-amplitude attenuation and a sub-millisecond peak-time displacement that is absorbed by the valley-to-peak detector's counter-based window. NeuroSEED is therefore positioned for on-chip detection and coarse classification rather than waveform-perfect offline sorting; for higher-fidelity sorting, the per-channel rate can be raised to 20.8 kHz by halving N to 16, at a 2× cost in normalised AFE area.

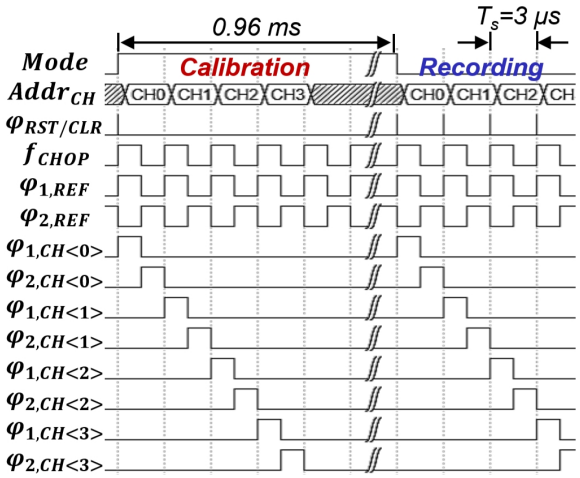


Fig. 5: Timing diagram illustrating the control signals for the analog multiplexer and chopper during the offset calibration and time-multiplexed recording modes.

The architecture is intended to scale to thousands of channels. *(i) Settling:* from (1), $f_{BW} \propto N \cdot f_{ch}/|\ln \epsilon|$; the present N=32 tile requires a minimum of 0.37 MHz at 10-bit settling (the implemented 700 kHz provides margin for WIS filtering), and a 1024-channel system is better realised as a 32×32 tile array than by pushing N beyond the range validated in Fig. 4. *(ii) Noise:* thermal noise scales as $\sqrt{f_{BW}}$, while MUX-Chop and WIS (see the following sub-section) suppress $1/f$ and folding contributions independently of N; the tiling path therefore preserves the per-tile noise budget. *(iii) MUX overhead:* the combined residual from R_{on}-limited settling, dummy-cancelled charge injection, and common-mode-rejected clock feedthrough remains well below the electrode-referred spike amplitude for $N \leq 64$, rather than growing unboundedly with N. *(iv) Spike fidelity:* because spikes are declared on a *relative* valley-to-peak excursion, static crosstalk does not create false events, and dynamic crosstalk is nulled by the WIS notch at the chopping frequency—making the tiled architecture robust at high channel density.

2) Circuit Implementation: The AFE topology, highlighted in green in Fig. 3, comprises a 32-channel multiplexed chopper, a capacitively coupled instrumentation amplifier (CCIA) with an impedance-boosting loop, and a tunable transconductor integrated with an ASC module for second-stage amplification, digitization, and offset calibration.

To mitigate the increased worst-case $1/f$ noise corner frequency due to multiplexing, chopper stabilization is employed, up-modulating neural signals beyond the flicker-noise corner [24]. To minimize switching complexity and suppress clock feedthrough, a 4-channel multiplexed chopper (4-Ch Mux-Chop) is placed after the analog multiplexer. The 32 channels are organized into eight groups, each sequentially routed through the chopping stage before amplification. Fig. 5 shows the timing diagram illustrating the interaction between the 4-Ch Mux-Chop and the analog multiplexer.

In practice, the MOS switches in the multiplexer and chopper exhibit non-ideal behaviors, including finite and signal-

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dependent on-resistance, charge injection, and leakage current. The finite $R_{\rm on}$ forms an RC network with the input capacitance, introducing amplitude attenuation at high multiplexing speeds and contributing kT/C thermal noise. Charge injection and clock feedthrough from control signals cause transient voltage spikes and baseline shifts, while leakage currents at high-impedance nodes lead to offset drift. These effects are exacerbated in microvolt-level neural signals, potentially degrading spike detection fidelity. To alleviate these issues, transmission-gate structures with low $R_{\rm on}$ are used, dummy switches are added for charge-injection cancellation, and differential signaling mitigates common-mode feedthrough.

The CCIA offers programmable closed-loop gains of 50, 12.5, or 6.25, determined by the capacitance ratio $C_{IN}/C_{NFB} = 7.3\,\text{pF}/(147\,\text{fF} \times \{1, 4, 8\})$. A positive-feedback capacitor (C_{PFB}) partially compensates for input impedance reduction due to chopping, with $C_{PFB} = C_{NFB}$ ensuring stability. The low-noise amplifier (LNA) core adopts an inverter-based, current-reuse topology to enhance noise efficiency, achieving a NEF of 2.1, as shown in Fig. 6. Complementary input pairs operate in weak inversion ($g_m/I_D \approx 26$) and utilize thick-oxide transistors to minimize gate leakage. Cascode transistors enhance open-loop gain for accurate closed-loop operation while isolating input devices from the output, mitigating the Miller effect associated with C_{gd}.

To attenuate up-modulated flicker and folding noise, a charge-sampling G_m–C integrator follows the first stage. Fig. 7 illustrates the circuit implementation, timing diagram, and transfer function of the charge-sampling operation. The transconductor G_m operates in two configurable modes, with transconductance adjustable by a factor of 100. In recording mode, a high-g_m setting is activated, and integration capacitors are placed at the front end of each ASC, with charge-clear switches connected in parallel.

Windowed Integration Sampling (WIS), controlled by $\varphi_{SMP<31:0>}$ and φ_{CLR}, is applied during demultiplexing of the mixed neural signals. The WIS technique yields a sinc-shaped frequency response, with deep notches aligned to the chopping frequency, effectively suppressing residual ripple and out-of-band interference. In addition to its filtering property, WIS inherently provides extra gain, given by $G_m T_{SMP}/C_{INT}$, where T_{SMP} is the integration window, C_{INT} is the integration capacitor, and G_m is the active transconductance. This integration-based approach improves spike detectability by reducing high-frequency noise folding during the subsequent digitization stage.

The total input-referred noise at the CCIA input can be decomposed into four physically distinct contributors:

$$\overline{v^2_{n,\text{tot}}} = \overline{v^2_{n,\text{th}}} + \overline{v^2_{n,1/f}} + \overline{v^2_{n,\text{fold}}} + \overline{v^2_{n,\text{mux}}}. \quad (2)$$

Thermal noise from the inverter-based LNA input pair scales as $8kT/(3g_m)\cdot f_{BW,\text{AFE}}$. Because the AFE bandwidth must cover $N\cdot f_{ch}$, this is the dominant term in any TDM front-end, and sets the current-reuse / weak-inversion sizing of Fig. 6. *Flicker (1/f)* noise is up-modulated above the signal band by chopping at 333 kHz; the residual in-band contribution is limited by chopper non-idealities (switch offset mismatch, residual ripple) rather than by the raw $K_f/(C_{ox}WL)$ term.

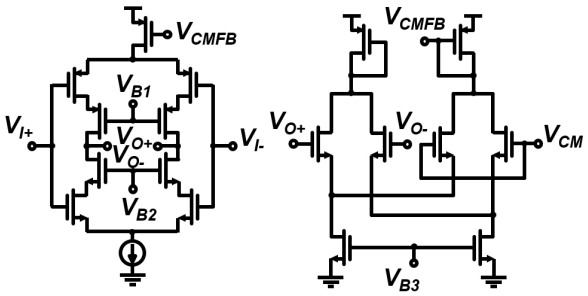


Fig. 6: Schematic of the inverter-based, current-reuse LNA core and its CMFB circuit.

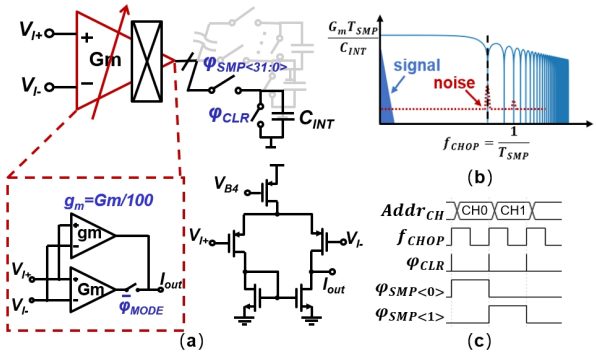


Fig. 7: The charge-sampling Gm-C integrator stage, showing (a) the dual-mode transconductor circuit, (b) its sinc-shaped filtering transfer function, and (c) the associated timing diagram.

Folding noise from wideband content aliased by the WIS sampler is suppressed by the sinc^2 notches at integer multiples of $1/T_{SMP}$, which align with the chopping frequency. *MUX-induced* noise comprises kT/C at the chopper switches, $R_{\rm on}$ thermal noise, and residual charge injection; the large $C_{IN} = 7.3$ pF sets a low kT/C floor, and the dummy-switch + differential topology further suppresses the injection component. After MUX-Chop+WIS the chain is thermal-noise-limited, consistent with the total 2.9 $\mu\text{V}_{\rm rms}$ IRN measured in Section IV.

B. Analog-to-Spike Converter and Event-Driven Spike Detection

1) Overview of Working Mechanisms: Following the shared analog front-end, a demultiplexer directs the time-interleaved neural signals to a bank of 32 parallel processing channels. Each channel contains an ASC and a spike detector that operate synergistically to perform neuromorphic data compression. This architecture leverages the inherent sparsity in neural activity; rather than sampling at a fixed rate, the ASC encodes the analog waveform into a stream of digital UP/DN events only when the signal amplitude changes significantly. This event-driven approach obviates the need to process or transmit quiescent periods, substantially reducing power consumption. As illustrated by the compression pipeline in Fig. 8, the process culminates in a hardware-efficient, integrate-and-fire algorithm. Detection is triggered upon identifying a signal

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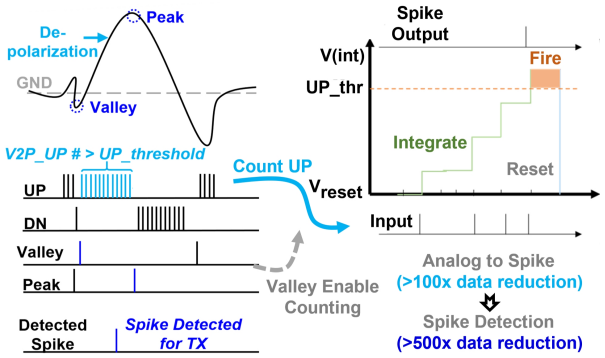


Fig. 8: The compression process from a neural signal to the output of the detected spike using the proposed event-driven spike detection scheme.

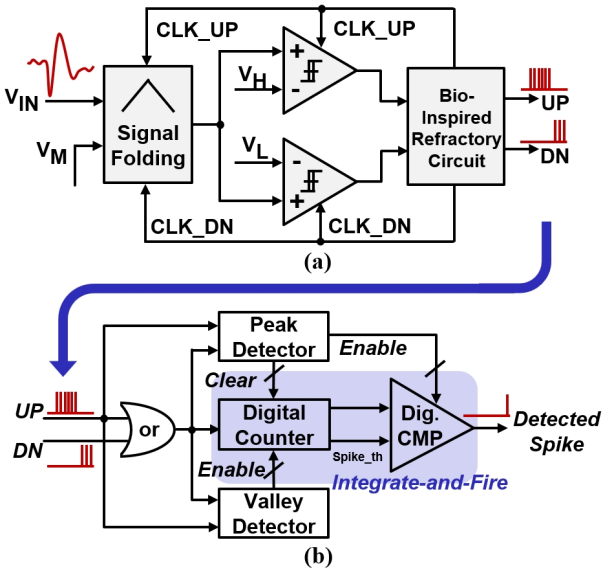


Fig. 9: The ASC and spike detector topologies.

valley, which prompts a counter to accumulate subsequent UP events. A spike is then declared if this accumulation, representing the valley-to-peak voltage swing, exceeds a programmable threshold, UP_threshold. Here, UP_threshold is a digitally tunable value, set via an external configuration register, which determines the minimum voltage excursion required to classify a signal fluctuation as a neural spike, thereby filtering out background noise.

The hardware implementation of the ASC and the asynchronous spike detector is detailed in Fig. 9. The ASC core (Fig. 9(a)) comprises three key sub-circuits: a 1-bit capacitive DAC that implements signal folding, two asynchronous comparators to detect threshold crossings (V_H and V_L), and a bio-inspired refractory circuit. Upon a threshold-crossing event, this refractory block generates the corresponding UP or DN data bit and a control clock (CLK_{UP}/CLK_{DN}). This clock serves a dual purpose: it triggers the signal-folding DAC to reset the input to V_M and enforces a refractory period that defines the ASC's maximum event rate.

The UP/DN event stream is then processed by the clockless spike detector shown in Fig. 9(b). Drawing inspiration from neuronal dynamics, this module operates asynchronously to minimize power. It enhances noise resilience by utilizing the relative valley-to-peak amplitude for spike detection rather than a fixed absolute threshold. The detector's internal logic, built around a valley/peak tracker, a digital counter, and a comparator, efficiently sifts through the ASC event stream to identify valid spikes. This event-driven processing chain, from analog input to spike output, not only curtails the data rate but also achieves a compact footprint. The spike detector itself occupies a mere 0.0007 mm^2 and consumes only 0.48 μW, highlighting the architecture's suitability for scalable, high-density neural recording systems.

Relative-excursion detection is intrinsically immune to slow baseline wander and to the residual DC offset after the servo loop. Three regimes warrant discussion. *(i) Overlapping spikes* within one refractory window are merged into a single event: this underestimates the spike count rather than producing false positives, which is the preferred failure mode for on-chip detection. *(ii) Non-stationary noise and artifacts*: because the detector tracks both endpoints of the excursion, slow baseline drift leaves the valley-to-peak difference invariant; motion artifacts that exceed the AFE dynamic range are clipped at the amplifier output and therefore propagate to the detector only as bounded excursions, limiting their contribution to the false-positive rate. *(iii) Refractory-controlled FP/FN trade-off*: the reconfigurable ASC refractory and detector blanking jointly tune the false-positive/false-negative balance, letting the detector be biased toward either fewer missed bursts or tighter false-positive suppression depending on the application; the operating point used for the in-vivo validation in Section IV is chosen so that moderate threshold mis-calibration does not destabilise the detection behaviour.

2) Design Consideration and Circuit Implementation: The circuit-level implementation of the ASC and the spike detector is detailed in Figure 10, with its design philosophy targeting minimal power and area under the constraints of asynchronous, event-driven operation. A primary challenge is ensuring the circuit's response time is sufficient for the fastest expected neural dynamics. The event-driven nature of the ASC means its instantaneous sampling rate is not constant, but rather adapts to the input signal's rate of change. For instance, a sinusoidal signal exhibits its lowest rate of change at its peaks and valleys, requiring fewer sampling events, while its highest rate of change occurs at the zero-crossings. Therefore, the circuit's speed must be sufficient to handle this maximum slew rate, which defines the minimum possible time between two consecutive sampling events, T_{min}. For a given equivalent resolution M and maximum input frequency f_{IN}, this relationship is:

$$T_{\min} = \frac{V_{\mathrm{LSB}}}{\left|\frac{dV(t)}{dt}\right|_{\max}} = \frac{1}{2^M \pi f IN} \tag{3}$$

For our target of a 6-bit resolution and a 10 kHz input bandwidth, the required circuit propagation delay is under 498 ns [38], a critical timing budget that governed the subsequent design choices.

The ASC resolution is jointly bounded by the V_{LSB} of the capacitive folding DAC (set by C_1/C_2 matching and the V_M

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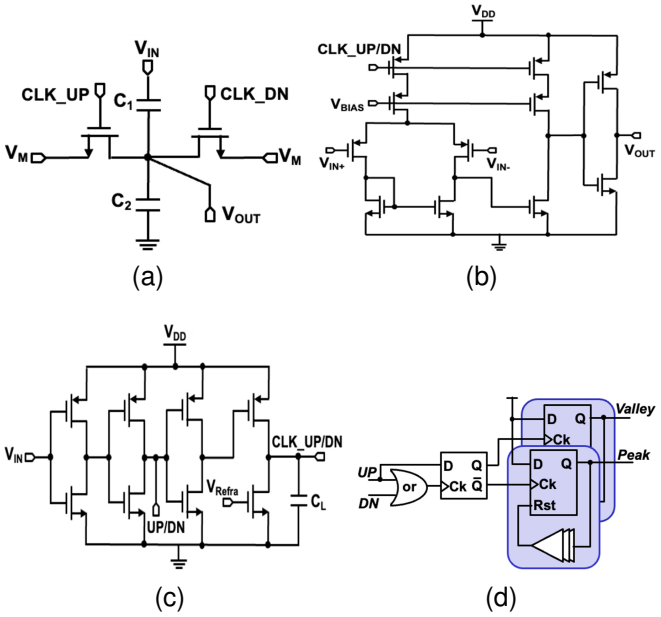


Fig. 10: Circuit diagrams of: (a) signal folding circuit, (b) asynchronous comparator, (c) refractory circuit, (d) peak and valley detector.

reference) and by the offset/hysteresis of the asynchronous comparators, together yielding the peak 7.6-bit ENOB of Fig. 16. Consistent with that figure, the low-frequency SNDR roll-off is dominated by these static precision effects, while the high-frequency roll-off reflects the finite loop delay $T_{\min}$, which truncates the maximum achievable event rate. The reconfigurable refractory circuit (Fig. 10(c)) is tuned so that its dead-time-limited event ceiling $1/T_{\text{refr}}$ exceeds the instantaneous event rate produced by a physiological neural action potential at the post-CCIA amplitude and slew rate; refractory-induced missed transitions and event saturation were accordingly absent across the in-vivo dataset. High-slew transients such as stimulation artifacts are clipped at the ASC input rather than mis-triggering the detector, because the downstream valley-to-peak detector requires a matched valley/peak pair.

To meet this speed requirement while adhering to a strict power budget, the ASC's core (Fig. 10(a)) employs a 1-bit switched-capacitor DAC for signal folding and digitization. This block translates the continuous analog input into a discrete UP/DN event stream by resetting the input baseline to a stable voltage, V_M, via charge redistribution upon being triggered. A key design trade-off between mismatch performance, area, and settling time led to the selection of 6 pF for the matched capacitors (C_1, C_2), while the NMOS switches were sized at 440 nm/220 nm (W/L) to achieve low on-resistance without excessive charge leakage. The event triggers that drive this DAC are generated by a 3-stage asynchronous comparator (Fig. 10(b)) and a reconfigurable refractory circuit (Fig. 10(c)). A system-wide low-power strategy was adopted, utilizing low-threshold-voltage (low-V_t) devices for reduced supply voltage operation. Furthermore, the comparators are aggressively power-gated via the refractory circuit's output during the signal folding phase, eliminating their static power consumption. The refractory circuit itself, realized as a pMOS stage with an adjustable nMOS load, provides a tunable "dead time" after each event. By modulating this load, the refractory period and thus the maximum event rate can be dynamically adapted to neural firing statistics. Finally, the resulting event stream is processed by a peak and valley detector (Fig. 10(d)), based on the topology in [39], to realize the spike detection algorithm.

C. Offset Calibration Feedback Loop

In multiplexed neural signal acquisition, the equilibrium potential at each electrode–tissue interface is determined by electrochemical interactions, ion concentration gradients, and electrode surface properties. These equilibrium potentials can vary by tens of millivolts across channels, and when referred to a common reference electrode, they appear as DC offsets superimposed on the neural signal. Unlike low-frequency neural activity, these offsets are quasi-static on short time scales but may drift over seconds to minutes due to physiological changes or electrode polarization.

In conventional neural recording front-ends, a high-pass filter (HPF) placed before the amplifier or at the input of each pixel can suppress such offsets. However, in a time-multiplexed architecture, the multiplexer operates before the main amplification, up-modulating the offsets to frequencies around the multiplexing rate. In this case, simple HPF removal becomes ineffective because the offset manifests as a step change at each channel switch, producing transient settling errors rather than a steady baseline shift.

An alternative approach is to use a high-resolution ADC to directly digitize both the large offsets and the small action potentials (APs). However, this forces the ADC dynamic range to be dominated by the offset, wasting resolution on the baseline and demanding a larger full-scale range, higher quantization bits, and increased sampling capacitance—resulting in significant area and power penalties.

To address these challenges, the proposed design incorporates an offset calibration loop integrated with a dual-mode event generator. This approach mitigates the risk of amplifier saturation while leveraging the offset-insensitive property of the asynchronous successive approximation converter by ASC.

1) Principle of Operation: The core concept is to implement a digital DC servo loop, which estimates and subtracts the DC offset before it reaches the LNA's gain-limited stage. The calibration phase proceeds as follows:

1) **Calibration Mode Activation** – The second-stage transconductor operates at a reduced transconductance ($g_m/100$), while all sampling switches are closed. This increases the effective integration time constant

$$\tau = \frac{C_{\text{eff}}}{g_m} \tag{4}$$

by a factor of ~ 3200 compared to recording mode, creating a very low cutoff frequency that averages out neural signal fluctuations and isolates the DC offset.

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2) **Offset Measurement** – The ASC integrates the reduced-bandwidth output and converts it into discrete events (up/down spikes) proportional to the sign and magnitude of the offset drift caused by changes in V_{CM}.
3) **Digital Accumulation** – A dual-counter accumulates the up/down events over a fixed observation window, producing a 6-bit signed digital word representing the average offset over 32 channels.
4) **DAC Feedback Injection** – The offset estimate drives a 6-bit capacitive DAC connected at the input, which applies an equal and opposite DC voltage to cancel the measured offset. The capacitive injection method ensures no additional noise path from the DAC reference supply and preserves the high input impedance of the CCIA.

The digital DC servo loop prevents amplifier saturation and optimizes both area and power efficiency. The calibration achieves a settling time of 0.96 ms and is performed every 4.8 s to accommodate slow offset drifts. After calibration, the neural interface transitions to signal monitoring mode, employing a closed-loop programmable-gain low-noise amplifier (LNA) with a current-reuse topology for efficient neural signal acquisition. This architecture assumes that the DC offset, while potentially large, is relatively uniform across closely spaced electrodes within a high-density array, or that the individual variations fall within the input dynamic range of the LNA after the average offset is removed. This design choice balances complexity with the constraints of high-density recording geometry.

2) Closed-Loop Transfer Function Analysis: The offset calibration feedback loop can be modeled as a discrete-time negative feedback system operating with a periodic update interval T_{update}. Denote the true offset drift as $D(z)$ and the input-referred offset after calibration as $V_{\text{off}}(z)$. The DAC applies a compensation voltage proportional to the accumulated ASC output, scaled by the loop gain β.

The discrete-time loop equation can be written as:

$$V_{\text{off}}[n+1] = (1-\beta)V_{\text{off}}[n] + D[n], \tag{5}$$

which in the z-domain becomes:

$$zV_{\text{off}}(z) = (1-\beta)V_{\text{off}}(z) + D(z). \tag{6}$$

Rearranging, we obtain the closed-loop transfer function from the drift input to the residual offset:

$$H(z) \triangleq \frac{V_{\text{off}}(z)}{D(z)} = \frac{1}{z-(1-\beta)}. \tag{7}$$

For a step-like DC drift (low frequency, $z \approx 1$), the DC gain reduces to:

$$|H(1)| = \frac{1}{\beta}, \tag{8}$$

showing that larger loop gain β yields better DC suppression.

The equivalent continuous-time pole location is given by:

$$s_p \approx \frac{\ln(1-\beta)}{T_{\text{update}}}, \tag{9}$$

so the closed-loop time constant is:

$$\tau_{\text{loop}} \approx -\frac{T_{\text{update}}}{\ln(1-\beta)}. \tag{10}$$

By designing β such that $\tau_{\text{loop}} \gg 1/f_{\text{AP,low}}$ (where $f_{\text{AP,low}} \approx 300\,\text{Hz}$), the loop exclusively tracks slow electrode drift without corrupting neural activity in the band of interest.

Finally, quantization in the DAC imposes a lower bound on the residual offset:

$$|V_{\text{res}}| \leq \frac{\Delta V_{\text{DAC}}}{2} = \frac{V_{\text{FS}}}{2^{N+1}}, \tag{11}$$

where V_{FS} is the DAC full-scale range and N is the DAC resolution (bits). For $V_{\text{FS}} = \pm 50\,\text{mV}$ and $N = 6$, this yields $|V_{\text{res}}| \leq 0.78\,\text{mV}$.

The shared calibration word implements a *common-mode* offset removal. Published characterisations of dense MEAs report per-array standard deviations of 3–10 mV [40], [41], dominated by slow common-mode drift rather than per-site randomness. Letting σ_{off} denote the per-channel spread, the worst-case post-calibration residual is:

$$|V_{\text{res}}|_{\text{wc}} \leq \underbrace{3\sigma_{\text{off}}}_{\text{diff.}} + \underbrace{\left(\frac{V_{\text{FS}}}{2^{N+1}} + |\Delta V_{\text{comp}}|\right)}_{\text{CM residual}}, \tag{12}$$

where the first term is the uncancelled inter-channel spread and the grouped second term combines DAC quantisation ($V_{\text{FS}}/2^{N+1} = 0.78$ mV for $V_{\text{FS}} = \pm 50$ mV, $N = 6$) with the residual comparator offset (<0.6 mV). For $\sigma_{\text{off}} = 5$ mV (mid-range of published values), $|V_{\text{res}}|_{\text{wc}} < 16.4$ mV, within the ± 25 mV CCIA saturation margin at 58.3 dB gain. Larger σ_{off} can be handled by partitioning the 32 channels into sub-groups with separate DAC codes—an extension not needed for the in-vivo dataset evaluated here.

D. Wireless Data Transmission

To alleviate the bandwidth bottlenecks common in high-density neural recording, the NeuroSEED system integrates an event-driven wireless transmitter. We employ an Impulse-Radio Ultra-Wideband (IR-UWB) architecture, chosen for its simplified hardware complexity and low energy-per-bit metric. The transmitter operates in a burst-mode topology: it remains in a deep sleep state and is only activated by the digital pulse outputs from the spike detector array.

As depicted in Fig. 11, the transmitter chain comprises a tunable edge-combiner impulse generator, a start-up oscillator, and a digital power amplifier (DPA). Upon receiving a 'Detected Spike' flag from the serializer, the impulse generator creates a narrow enable window. This signal gates the LC-tank oscillator, which is designed with a current-reuse complementary structure to maximize voltage swing while operating under a reduced supply voltage. The oscillating signal is then buffered and driven to the antenna via the DPA. The DPA utilizes a push-pull inverter-based output stage, ensuring efficient driving of the 50 Ω antenna load while providing isolation for the oscillator.

The modulation scheme is Pulse Position Modulation (PPM) compatible, though in this specific implementation, On-Off

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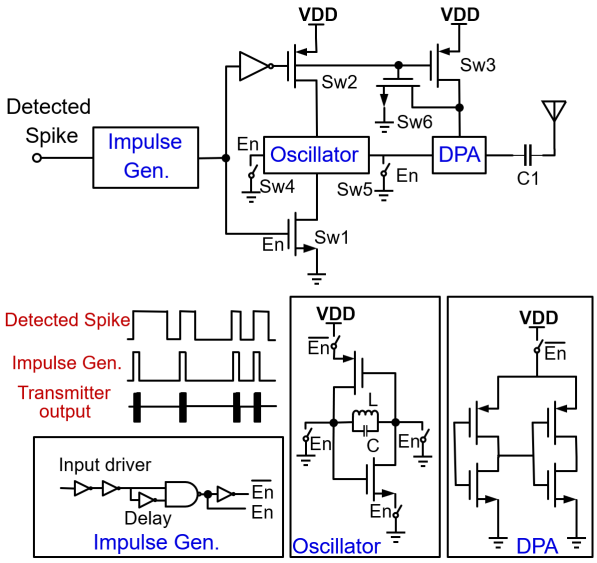


Fig. 11: Architecture and circuit diagrams of the proposed IR-UWB.

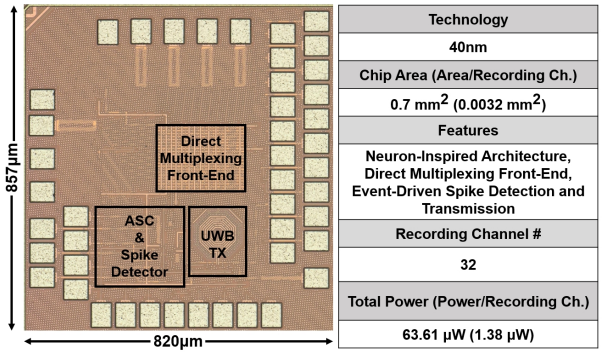


Fig. 12: Chip micrograph and performance summary.

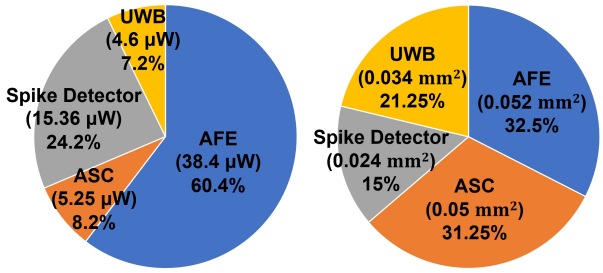


Fig. 13: Power and area breakdown of the proposed chip.

Keying (OOK) is inherently utilized due to the spike-triggered nature of the data. This 'send-on-event' strategy ensures that power is consumed only during the transmission of valid neural information, resulting in a system power that scales linearly with the biological firing rate.

The link budget for evaluating power requirements and signal quality of the wireless transmitter is summarised as follows. The average transmitted power P_t is derived from the measured PSD of -42.05 dBm/MHz over a 510 MHz bandwidth, giving $P_t \approx -14.9$ dBm ($P_t = -42.05 + 10\log_{10}(\mathrm{BW})$). This level is 0.75 dB below the FCC spectral mask limit of -41.3 dBm/MHz and remains orders of magnitude lower than the IEEE C95.1 specific absorption rate (SAR) safety limits [42]. The channel loss is primarily governed by two components: free-space path loss and in-tissue attenuation. Based on published FDTD simulations and experimental measurements at 4 GHz, the combined attenuation is estimated to be in the range of 30–40 dB for a typical 15 mm cranial propagation path (brain, skull, and skin). The proposed IR-UWB transmitter is evaluated using a noncoherent receiver [43], achieving a BER of 10^{-3} for a maximum data rate of 100 Mb/s over a communication link consisting of 1.5 cm tissue depth and 3 cm free-space transmission. Notably, the event-triggered transmission scheme incorporates a data-reduction mechanism that lowers the pulse repetition frequency, effectively improving the link budget. This allows the use of higher-energy pulses without violating the FCC spectral mask constraints. As a result, the received signal strength increases, providing additional link-budget margin and improved BER compared to a continuously transmitting baseline.

IV. Measurement Results

The proposed SoC was fabricated in a TSMC 40-nm CMOS process with a die area of 857 μm $\times$ 820 μm. The chip micrograph and key performance specifications are shown in Fig. 12, while Fig. 13 summarizes the power and area distribution among the major building blocks. The total active area is 0.16 mm^2, and the system consumes 63.61 μW from a 1.2-V supply.

A. Electrical Characterization

1) Direct Multiplexing Front-End: The performance of the direct-multiplexing front-end was experimentally evaluated. Fig. 14 shows the measured frequency response under different programmable gain settings. The low-frequency gain can be adjusted from 41.2 dB to 58.3 dB, while maintaining a stable mid-band response, thereby providing flexible amplification for a wide range of input levels.

Fig. 15 presents the measured output spectrum with a 1-mV$_{\mathrm{pp}}$ input sinusoid at 200 Hz. The circuit achieves an SNDR of 40.3 dB and an SFDR of 51.3 dB. The input-referred noise (IRN), integrated over the 1–1 kHz bandwidth, is 2.9 μV$_{\mathrm{rms}}$. These results demonstrate that the proposed front-end provides sufficient dynamic range and low-noise performance for neural recording applications.

2) Analog-to-Spike Converter: The dynamic performance of the ASC was characterized by measuring the SNDR of a reconstructed sinusoid across a frequency range of 100 Hz to 10 kHz, as presented in Fig. 16. The ASC achieves a peak SNDR of 47.2 dB, which corresponds to a 7.6-bit ENOB, at an input frequency of 500 Hz. The performance roll-off below this peak is attributed to the static precision limitations of the signal folding circuit and comparators. Conversely, as the input frequency increases beyond 1 kHz, the SNDR becomes constrained by the finite loop delay inherent in the event-driven architecture, eventually settling to a relatively stable level.

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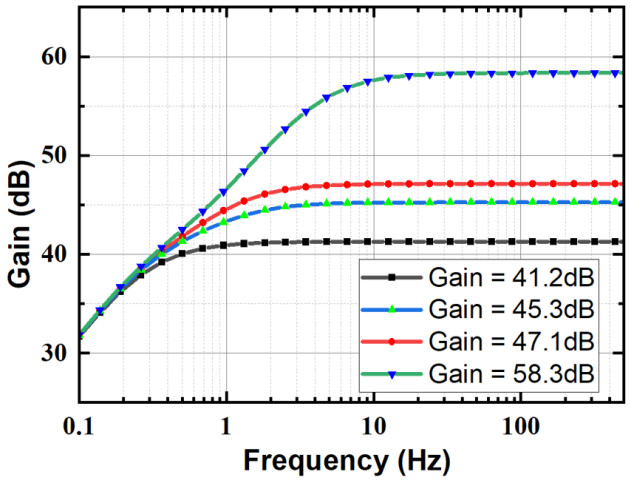


Fig. 14: Measured frequency response of the direct-multiplexing front-end under different gain settings.

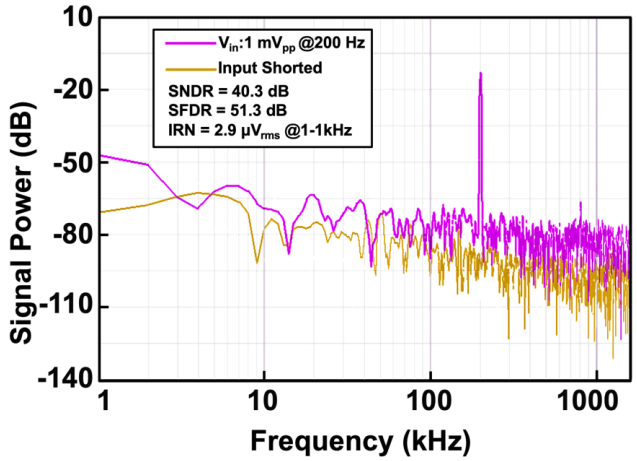


Fig. 15: Measured output spectrum of the direct-multiplexing front-end with a 1-mV_{pp} sinusoidal input at 200 Hz.

3) IR-UWB: Fig. 17 illustrates the measurement results obtained from the fabricated ultra-wideband transmitter. To evaluate its time-domain characteristics, a pseudo-random binary sequence (PRBS) at a data rate of 100 MHz, generated by a Keysight arbitrary function generator, is used as the modulated input signal. Furthermore, the transmitter's spectral performance is analyzed by measuring its power spectral density (PSD) with a Keysight N9020B spectrum analyzer.

The measured PSD reveals that the transmitter operates effectively with a center frequency of 4.25 GHz and a bandwidth of 510 MHz. The maximum recorded PSD is -42.05 dBm/MHz, ensuring compliance with Federal Communications Commission (FCC) regulatory standards for ultrawide band transmission.

B. Validation with In-Vivo Dataset

To validate the proposed system, an in-vivo rat neural signal dataset from [44] was employed. The dataset was recorded from an adult female Sprague Dawley rat using a custom-made

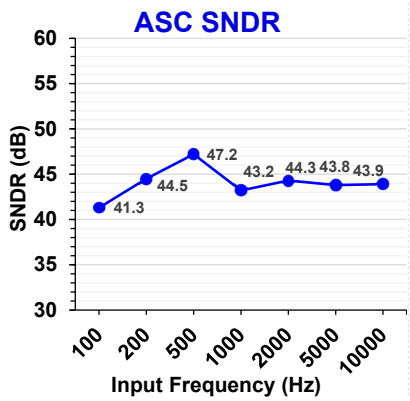


Fig. 16: ASC SNDR against input sinewave signal frequency ranging from 100 Hz to 10 kHz.

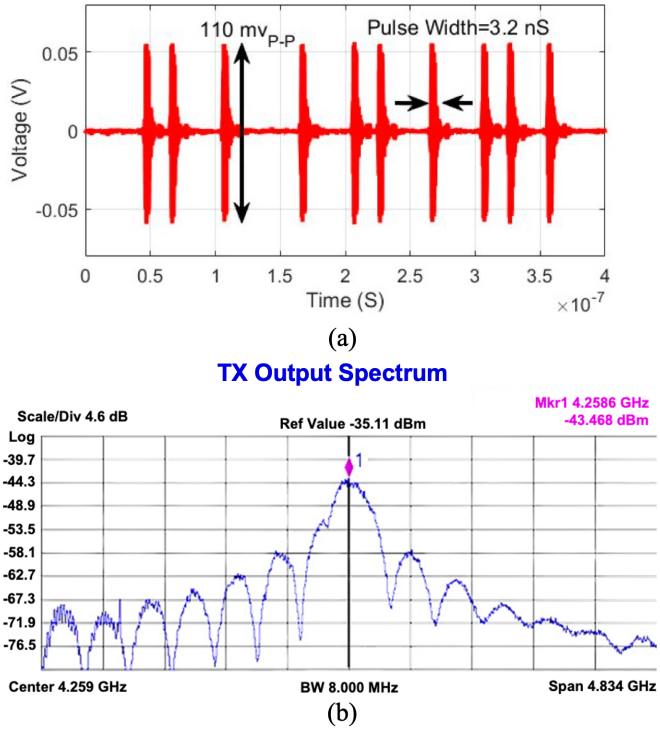


Fig. 17: Measurement results of the transmitter: (a) emitted signal in time domain, (b)PSD (dBm/MHz).

electrode array attached to a teased fascicle of the L5 dorsal rootlet. It contains 20 recordings (10 at rest and 10 during dermatome stimulation), each consisting of five channels of amplified and digitized raw voltage traces.

Fig. 18 shows the transient AFE output obtained with the in-vivo dataset, while Fig. 19 illustrates the measured event-driven spike detection. These results confirm that the proposed SoC can reliably process real neural signals and achieve effective data reduction through event-driven encoding.

Fig. 19 shows that the on-chip valley-to-peak detector reliably tracks the annotated spike locations throughout the recording. The relative-amplitude criterion confers two architectural advantages over absolute-threshold schemes: slow baseline drift leaves the valley-to-peak excursion invariant

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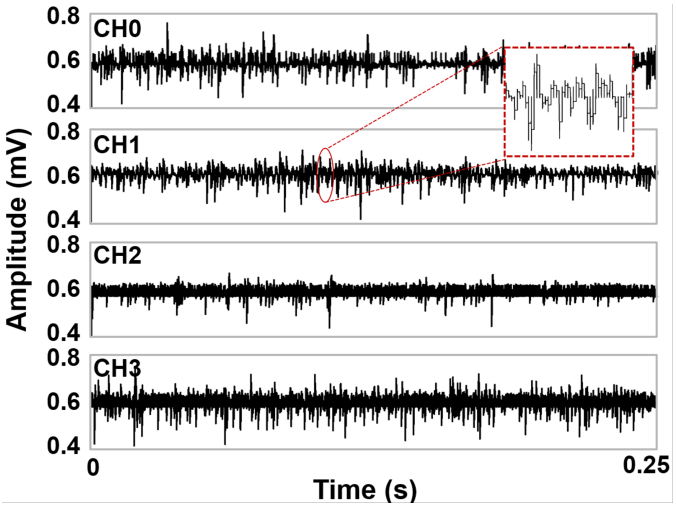


Fig. 18: AFE output given in-vivo dataset.

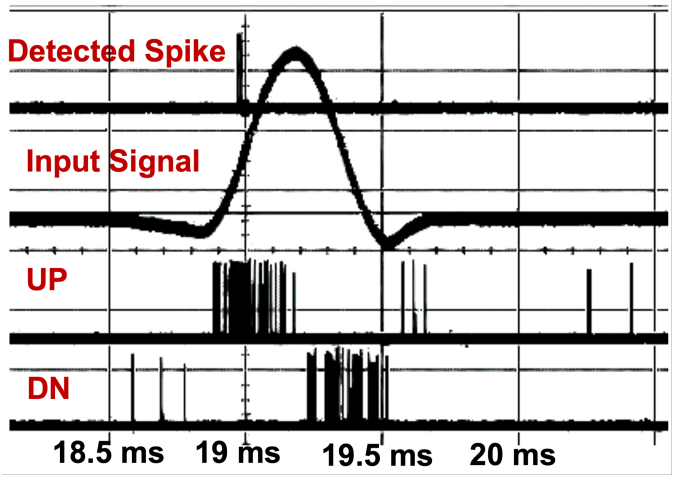


Fig. 19: Measured event-driven spike detection using in-vivo dataset.

(Section III-B), and the detection threshold is set on a per-event excursion rather than on an absolute level, so slow impedance drift and reference-electrode shifts do not require recalibration. A head-to-head quantitative benchmarking against threshold-based and Teager-energy-operator baselines on public datasets is outside the scope of this chip-level paper and is left to future algorithm-level work, as the present focus is on the circuit co-design that makes on-chip valley-to-peak detection feasible within the TDM + ASC constraints.

The $>500\times$ data-reduction figure is the ratio of an equivalent Nyquist-rate payload to the aggregate spike-gated payload produced by the chip on the in-vivo dataset. For the Nyquist baseline we take the 32 channels, the per-channel sampling rate $f_s = 10.4$ kHz, and a 10-bit equivalent sample depth sufficient to observe the AFE output, giving $32{\cdot}10.4$ kHz$\cdot$10 bit $\approx$ 3.3 Mbps. The spike-gated payload is the aggregate bit rate that the IR-UWB transmitter emits while operating on the dataset, with physical-layer framing overhead excluded on both sides of the ratio. Higher-resolution Nyquist baselines (e.g. 12-bit, 25 kHz) yield correspondingly larger reductions; adding physical-layer UWB framing to the denominator lowers the figure but keeps it above $400\times$.

The architecture is designed with chronic use in mind. (i) The $7\times$ programmable AFE gain range (41.2–58.3 dB) provides headroom for slow impedance-induced amplitude loss—with the caveat that noise also scales at the retuned point, so the end-to-end SNR benefit is restricted to the regime where amplitude loss dominates noise rise, typical of the $\sim 3\times$ impedance growth reported in the literature. (ii) The servo loop, re-armed every 4.8 s, tracks electrode-polarisation drift on seconds-to-minutes scales. (iii) The valley-to-peak detector is intrinsically insensitive to baseline wander and motion artifacts (Section III-B). A full chronic *in-vivo* characterisation is planned for follow-up work.

C. Discussion and Comparison With State-of-the-Art Works

Table I summarizes a comparison with representative state-of-the-art neural recording systems. Compared to prior works, this design achieves competitive noise efficiency and energy consumption while offering integrated data reduction capability through event-driven spike encoding. The overall system demonstrates a favorable trade-off among scalability, energy efficiency, and functionality, validating its potential for next-generation implantable neural interfaces.

Because Table I spans five CMOS nodes (22–180 nm), we report two node-agnostic metrics. The bandwidth-normalised per-channel power $\mathrm{FoM_{BW}} = P_{\mathrm{ch}}/\mathrm{BW_{eff}}$ gives 0.265 nW/Hz at $P_{\mathrm{ch}} = 1.38$ μW and 5.2 kHz, on the same order as 28 nm $\Delta\Sigma$/boxcar front-ends [17] ($\sim$0.057 nW/Hz for amplifier-only power) and $\sim 4\times$ better than 65/180 nm counterparts, despite NeuroSEED also providing on-chip spike detection. The effective transmitted-bit energy $E_{\mathrm{tx}} = P_{\mathrm{TX}}/(N_{\mathrm{ch}} r_{\mathrm{spike}} B_{\mathrm{pkt}}) \approx 0.78$ nJ/tx-bit captures the benefit of spike-gated telemetry once the much lower duty cycle is accounted for. The two metrics confirm that the architectural contributions—event-driven encoding, digital offset servo, and spike-gated telemetry—translate into system-level efficiency not attributable to process-node bias.

V. Conclusion

This paper has presented a 32-channel neuromorphic neural interface, effectively addressing critical challenges in implantable multichannel neural interface design, such as high power consumption, limited data transmission efficiency, and constrained chip area. By adopting biological neuron-inspired signaling principles, the proposed system incorporates several key innovations: an area-efficient, time-division multiplexed analog front-end integrated with digital servo loop offset compensation; an adaptive event-driven ASC for substantial data compression; a clockless spike detector with improved noise tolerance; and an IR-UWB transmitter activated selectively by neural spike events to minimize transmission power. Fabricated using advanced CMOS technology, the system achieved exceptional performance metrics, notably consuming only 1.38 μW per channel and 4.6 μW in total transmitter power during simultaneous full-channel operation. Experimental validation confirmed the effectiveness of the proposed architecture, demonstrating significant advancements in power efficiency,

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TABLE I: Comparison of Neural Recording Systems.

Parameter	**This Work**	**JSSC'25** [29]	**JSSC'25** [28]	**JSSC'25** [27]	**JSSC'24** [17]	**ISSCC'23** [45]	**JSSC'22** [46]
System							
Technology (nm)	40	180	180	65	28	65	22
Supply Voltage (V)	1.2/0.6	0.9/0.7	1.2	0.9/1.2/3.3	1.0	0.8	1.2
# Recording Channels	32	8	192	256	1024	128	256
Total Power (μW)	63.61	126	3650	803.8	508.7	1098	412.16
Chip Area (mm^2)	0.7	0.72	0.48	1.46	3.27	4	0.256
Total Power/Channel (μW)	1.99	14	19	3.14	0.496	17.2	1.61
Total Area/Channel (mm^2)	0.02	0.09	0.0025	0.0057	0.0032	0.031	0.001
Neural Recording							
Architecture	TDM + ASC[a]	CT-ΔΣ	OTA-C IΔΣ	TDM + SAR	Boxcar + SS	SC-DDC	I-ΔΣ
Power/Channel (μW)	1.38	14	9.62	2.9	0.268	9.87	1.61
Area/Channel (mm^2)	0.0032	0.09	0.00175	0.0057	0.00129	0.02	0.001
Bandwidth	5.2 kHz	10 kHz	10 kHz	10 kHz	0.3-5 kHz	10 kHz	1 kHz
Simultaneous Recording BW (kHz)[d]	166.5	80	1920	640	5120	1280	256
Avg. IRNoise ($nV/\sqrt{Hz}$)	91	60	112	48	104.65	209	69
Data TX							
Data Reduction	Yes	No	No	No	Yes	No	No
Data Rate	Neuron Firing Rate	-	-	-	1.12Mbps	20.16 Mbps	-
TX Power (μW)	4.6	-	-	-	-	16.1	-
Energy Eff. (pJ/bit)	17.16[d]	-	-	-	-	0.8	-

[a] Analog-to-Spike Converter.
[b] Only amplifier power included.
[c] Estimated from die photo.
[d] Simultaneous recording channels × BW. Measured at 25 Mbps.

compactness, and data handling capabilities compared to state-of-the-art systems. These outcomes highlight the proposed system's potential for practical implementation in high-density, long-term neural monitoring applications.

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